



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Jerrage

Product name: prochloraz 45% W/V EC

Product use: Fungicide

Relevant identified uses of the substance/mixture and uses advised against:

Used on cereals (foliar spray and seed treatment) and mushrooms, as proposed by the applicants.

2. HAZARD IDENTIFICATION

Classification of the substance or mixture

Acute aquatic toxicity: Category 2 – (H401)

Chronic aquatic toxicity: Category 1 – (H410)

Flammable liquids: Category 4 – (H227)

Label elements

Hazard pictograms:



Signal word:

warning

Hazard statements:

H401 – Toxic to aquatic life

H410 – Very toxic to aquatic life with long lasting effects

H227 – Combustible liquid

Precautionary statements:

P102 – Keep out of reach of children

P210 – Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P501 – Dispose of contents/container to an approved waste disposal plant.



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Other hazards

No information available.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: *N*-propyl-*N*-[2-(2,4,6-trichlorophenoxy)ethyl]imidazole-1- carboxamide

Common name: prochloraz

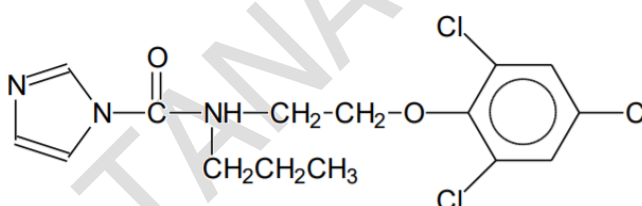
Ingredients and composition:

Component	CAS No.	% W/W
Active ingredient - Prochloraz	67747-09-5	38 - 42
Butan-1-ol	71-36-3	3 - 5
Butyrolactone	96-48-0	3 - 5

Formula: $C_{15}H_{16}Cl_3N_3O_2$

Molecular weight: 376.7 g/mol

Structural formula:



4. FIRST AID MEASURES

Description of first aid measures

General advice:

In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible). First aider: Pay attention to self-protection.

In case of swallowed:

Rinse mouth. Drink plenty of water. If symptoms persist, call a physician.

In case of eyes contact:

Immediately with soap and plenty of water while removing all contaminated clothes and shoes. Consult a physician if necessary.

In case of skin contact:

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Consult a physician if necessary.



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In case of Inhaled contact:

Remove to fresh air. If breathing is irregular or stopped, administer artificial respiration.

Call a physician.

Self-protection of the first aider:

Use personal protective equipment as required.

Most important symptoms and effects, both acute and delayed

Symptoms: None known

Indication of any immediate medical attention and special treatment needed

Note to physicians: Treat symptomatically

5. FIRE FIGHTING MEASURES

Flammability: Not applicable

Flash point: 82 °C

Suitable Extinguishing Media:

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media:

No information available.

Special hazards arising from the substance or mixture:

No specific hazard known.

Advice for firefighters:

In the event of fire, wear self-contained breathing apparatus

In the event of fire and/or explosion do not breathe fumes.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal precautions, protective equipment and emergency procedures

Personal precautions: Use personal protective equipment as required. Evacuate personal to safe areas. Keep people away from and upwind of spill/leak

For emergency responders: Use personal protection recommended in Section 8.

Environmental precautions:

Prevent entry into waterways, sewers, basements of confined areas. Do not flush into surface water or sanitary sewer system. Prevent further leakage or spillage if safe to do so. Prevent product from entering drains.

Methods and material for containment and cleaning up



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Methods for cleaning up:

Take up mechanically, placing in appropriate containers for disposal.

Reference to other sections

Other information:

See also section 8, 13

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Precaution for Safe Handling:

Advice on safe handling:

Use personal protective equipment as required. Avoid contact with skin, eyes or clothing. Wash contaminated clothing before reuse. Do not breathe dust/fume/gas/mist/vapor/spray. Do not eat, drink or smoke when using this product. Use with local exhaust ventilation.

General hygiene considerations: When using, do not eat, drink or smoke. Wash contaminated clothing before reuse. Regular cleaning of equipment, work area and clothing is recommended.

Conditions for safe storage, including any incompatibilities:

Storage conditions:

Keep container tightly closed in a dry and well-ventilated. Keep out of the reach of children. Keep containers tightly closed in a cool, well-ventilated place. Keep in properly labeled containers.

Specific and use(s):

Risk Management Methods (RMM): The information required is contained in this Safety Data Sheet.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

National occupational exposure limits

Chemical Name	European Union	United Kingdom	France	Spain	Germany
Butan-1-ol 71-36-3		STEL: 50 ppm STEL: 154 mg/m ³ skin	STEL: 50 ppm STEL: 150 mg/m ³	S* STEL: 50 ppm STEL: 154 mg/m ³	TWA: 100 ppm TWA: 310 mg/m ³ Ceiling/Peak: 100 ppm Ceiling/Peak: 310 mg/m ³
Butyrolactone 96-48-0					Skin
Chemical Name	Italy	Portugal	Netherlands	Finland	Denmark
Butan-1-ol 71-36-3		TWA: 20 ppm		TWA: 50 ppm TWA: 150 mg/m ³ STEL: 75 ppm	Ceiling: 50 ppm Ceiling: 150 mg/m ³ skin



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				STEL: 230 mg/m ³ skin	
Butyrolactone 96-48-0				TWA: 50 ppm TWA: 14 mg/m ³ STEL: 250 ppm STEL: 70 mg/m ³ skin	
Chemical Name	Austria	Switzerland	Poland	Norway	Ireland
Butan-1-ol 71-36-3	TWA: 200 ppm TWA: 600 mg/m ³ STEL: 50 ppm STEL: 150 mg/m ³	TWA: 50 ppm TWA: 150 mg/m ³ STEL: 50 ppm STEL: 150 mg/m ³	TWA: 150 mg/m ³ STEL: 50 mg/m ³ skin	Ceiling: 25 ppm Ceiling: 75 mg/m ³ skin	TWA: 20 ppm STEL: 25 ppm STEL: 75 mg/m ³ skin

Derive no effect level (DNEL)

No information available

Predicted no effect concentration (PNEC)

No information available

Exposure control

Engineering controls:

Ensure adequate ventilation, especially in confined areas.

Personal protective equipment

Eye/face protection:

Tight sealing safety goggles.

Body protection:

Suitable protective clothing, Apron, Gloves made of plastic or rubber.

General hygiene consideration:

When using do not eat, drink or smoke. Wash contaminated clothing before reuse. Regular cleaning of equipment, work area and clothing is recommended.

Environmental exposure controls:

Do not allow into any sewer, on the ground or into any body of water.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:

Liquid

Color:

Clear amber

Odor:

Characteristic

pH:

7 – 8 (CIPAC MT 75.3 solution 1%)

Flash point:

82 °C (EEC A.9)

Relative density:

1.130 – 1.135 (OECD 109)

Kinetic viscosity:

331 x 10⁻⁶ mm²/s (OECD 114 at 20°C)



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10. STABILITY AND REACTIVITY

Reactivity:	Not available
Chemical stability:	Stable under normal conditions
Possibility of hazardous reactions:	None under normal processing
Conditions to avoid:	Heat, flames and sparks
Incompatible materials:	No information available
Hazardous decomposition products:	None under normal use conditions.

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Tetraconazole:

LD50 (Oral, rat):	> 2,000 mg/kg
LD50 (Dermal, rat):	> 2,000 mg/kg
LC50 (Inhalation, rat, 4h):	> 4.99 mg/L air (max attainable conc.)
Skin irritation (rabbit):	Not irritant.
Eye irritation (rabbit):	Not irritant.
Skin sensitization (Guinea pig):	Not classified as a skin sensitizer

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, prochoraz:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h):	3.05 mg/L
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Invertebrates – Acute toxicity:

Crustacea (Daphnia magna),

EC50 (48 h):	4.0 mg/L
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Invertebrates – Acute toxicity:

Algae (Selenastrum capricornutum),

EC50 (72 h):	1.42 mg/L
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Birds – Toxicity:

Bobwhite quail,

LD50:	624 mg/kg
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Arthropod – Toxicity:

Bee (*Apis mellifera*),

LD50 (48 h): 684 ug/bee

Persistence and degradability

Abiotic degradation

Water,

DT50: 11.7 days

Soil,

DT50: 223.62 days

Biodegradation,

Not readily biodegradable

Bioaccumulative potential

Partition Coefficient (n-octanol/water),

Log Pow: 4.3 – at 20-25°C

Bioconcentration factor (ECF),

Prochloraz 1222-86554

Results of PBT and vPvB assessment

The components in this formulation do not meet the criteria for classification as PBT or vPvB

13. DISPOSAL CONSIDERATIONS

Waste treatment methods:

Waste from residue/unused products: Disposal shown in accordance with applicable regions national and local laws and regulations.'

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

IMDG/IMO

UN No: 3082

Dangerous Goods Class: 9

Packaging Group: III

Marine pollutant: Not regulated Yes

Proper shipping name: ENVIRONMENTALLY HAZARDOUS SSUBSTANCE, LIQUID
N.O.S. (Prochloraz)



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Environmental hazard: Not regulated Yes

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Packaging Group: III

Environmental hazard: Not regulated Yes

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N.O.S. (Prochloraz)



IBC Code:

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

16. OTHER INFORMATION

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.